Exp 8: Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

1. Aim

To configure User Management, custom organizational roles, hierarchical reporting relationships, permission profiles, and record-level Data Sharing Rules within the Zoho Creator application (*Vehicle Services & Repair Request*) to enforce Role-Based Access Control (RBAC) and least-privilege data governance across administrative and operational roles.

2. Procedure

Phase A: Access Centralized User Management & Permissions Console

Step 1: Logged into the Zoho Creator developer workspace and navigated to the *Vehicle Services & Repair Request* application.

Step 2: From the top primary navigation bar, selected **Settings** to open the administrative configuration console.

Step 3: Under the **PERMISSIONS** category in the left sidebar menu, clicked on **User Permissions** to load the centralized user access management dashboard (*"Add and manage users' permissions and roles in the application. Define the visibility of records for each users."*).

Phase B: Review Application Permission Sets and Access Profiles

Step 4: In the **Users** console, selected the **Permissions** tab to examine the application's component-level security profiles:

- **Read Profile:** Grants read-only viewing permissions across all application components (Forms, Reports, and Pages) without record mutation rights.
- **Write Profile:** Grants complete read and write access across all forms, workflows, and reports, permitting record creation, updating, and deletion.

Step 5: Inspected the **Add Permission** feature used to define custom granular permission sets specifying component-level CRUD (Create, Read, Update, Delete) rights.

Phase C: Establish Organizational Role Hierarchy Tree

Step 6: Switched to the **Roles** tab within the Users management console to evaluate and structure the organizational hierarchy tree, starting from the pre-configured root node `CEO`.

Step 7: Clicked the blue **Add Role** button to open the role definition drawer:

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- **Role Name:** Entered Service Manager.
- **Reports to:** Selected CEO from the hierarchical reporting dropdown list.
- **Peer Sharing:** Evaluated the *Share Data with Peers* option (left unchecked to enforce vertical departmental confidentiality).
- Clicked **Save** to commit the managerial role to the hierarchy.

Step 8: Repeated the role provisioning procedure to create the operational field role Service Technician, configuring its reporting line directly under CEO.

Step 9: Expanded the organizational role tree to confirm that both Service Manager and Service Technician are correctly situated within the reporting structure.

Phase D: Configure Granular Record-Level Data Sharing Rules

Step 10: Switched to the **Data Sharing** tab to implement record-level visibility boundaries, ensuring users only access records pertinent to their organizational responsibilities.

Step 11: Clicked **New Share Rule** (or selected the rule editor drawer) to configure a custom data sharing policy:

- **Rule Name:** Entered Technician Read Only Rule.
- **Target Form / Report:** Selected Service_Request_Form Report as the primary operational data entity.
- **Records shared from:** Selected Roles and Subordinates > Service Technician to capture all vehicle service records generated or handled by technicians.
- **Records shared To:** Selected Roles > CEO to grant organizational executive visibility over technician records.
- **Level of access:** Selected Read Only to preserve record integrity and prevent unintended executive modification of operational logs.

Step 12: Clicked **Save** to enforce the record sharing rule across the Zoho Creator database engine.

Phase E: Verification and Audit of Enforced Security Policies

Step 13: Verified the active record sharing policy in the Data Sharing summary table, confirming that rule parameters were successfully deployed and active with Source set to R:Service Technician, Destination set to R:CEO, Target Entity set to Service_Request_Form Report, and Access Level set to Read Only.

3. Output

3.1 User Management Console — Permissions Tab and Access Profiles

The centralized Users settings interface displaying the default component permission profiles (*Read* and *Write*) and the *Add Permission* action button for custom profile provisioning:

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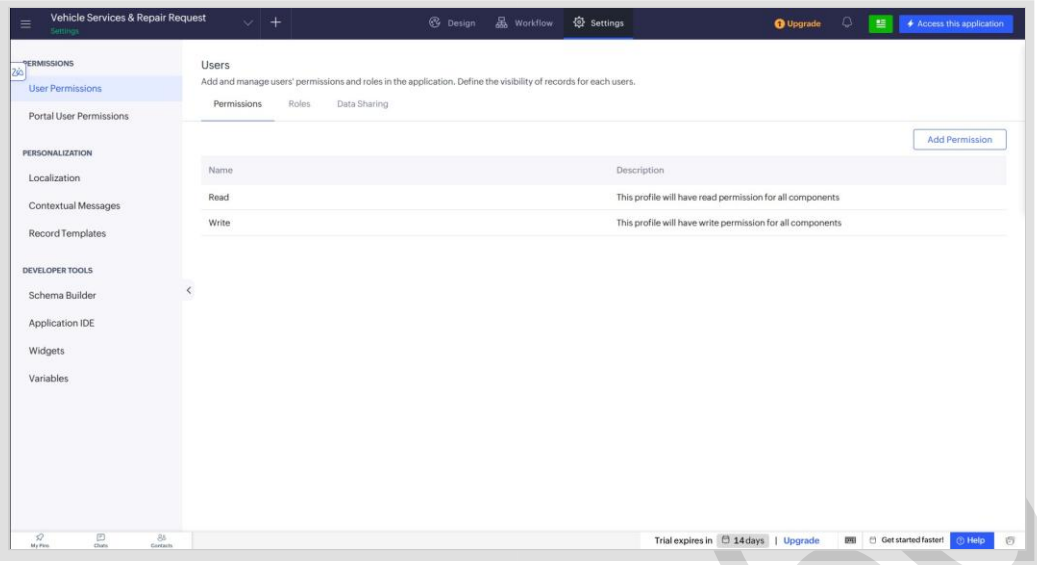


Figure 1: Zoho Creator User Management Dashboard — Reviewing Default Permission Profiles (Read & Write) under Settings > User Permissions

3.2 Roles Architecture Overview — Root Hierarchy Tree

The Roles tab within the User Management console displaying the initial single-node root organization hierarchy tree under the *CEO* role:

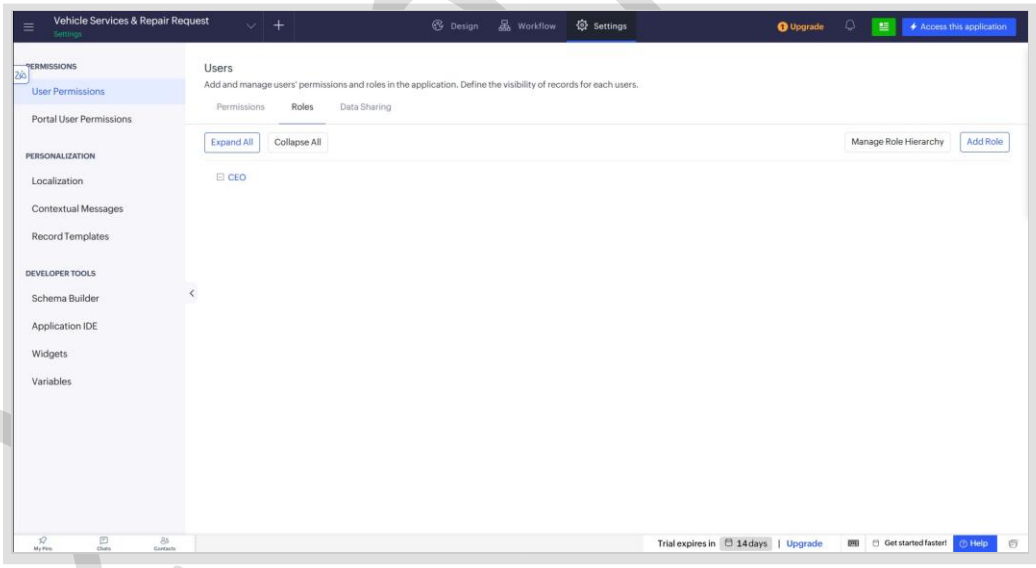


Figure 2: Organizational Roles Overview — Root Hierarchy Tree Anchored at CEO with Action Controls (Expand/Collapse All, Manage Role Hierarchy, Add Role)

3.3 Role Provisioning Drawer — Adding "Service Manager" Role

The *Add Role* slide-out drawer specifying role properties, including Role Name (Service Manager), supervisory reporting relationship (Reports to: CEO), and peer data-sharing flags:

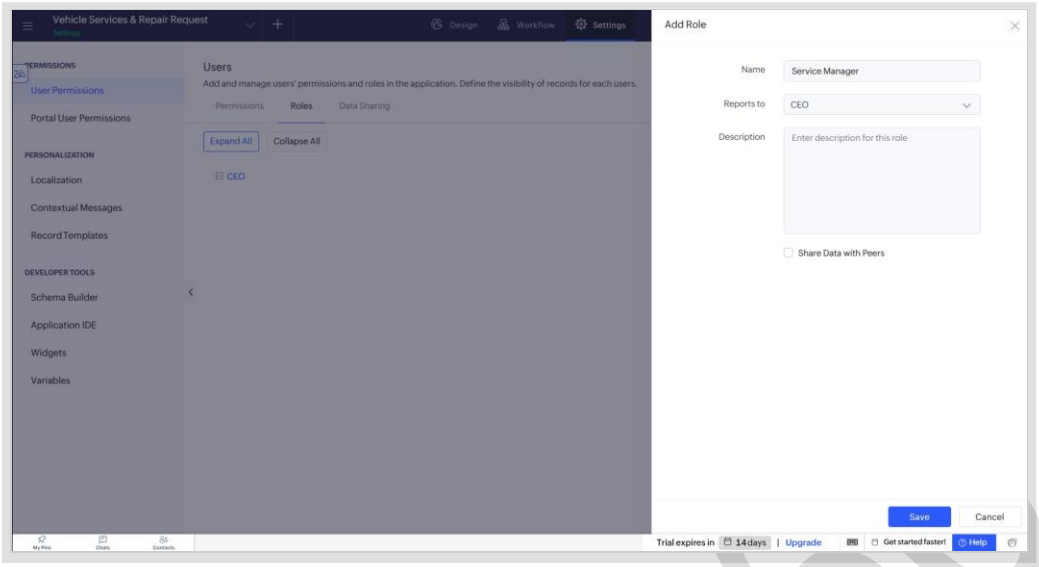


Figure 3: Role Provisioning Configuration Drawer — Defining "Service Manager" Subordinate to "CEO"

3.4 Complete Role Hierarchy Architecture — Multi-Tier Organization Tree

The expanded Roles hierarchy tree displaying the structured organizational breakdown with *Service Manager* and *Service Technician* operating beneath executive management:

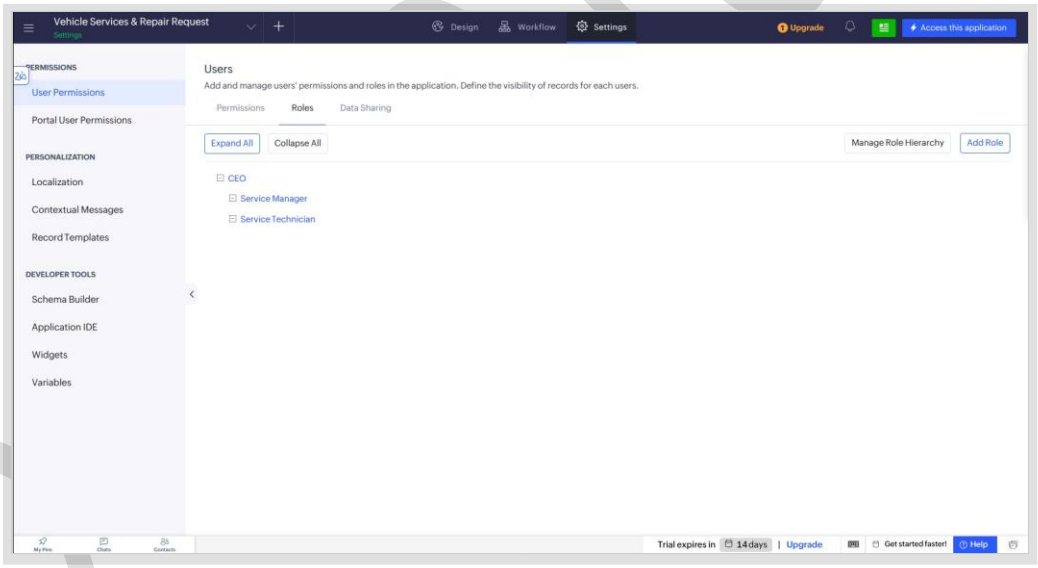


Figure 4: Configured Multi-Tier Organizational Role Hierarchy — CEO > Service Manager & Service Technician

3.5 Data Sharing Rule Builder — Configuring "Technician Read Only Rule"

The *Edit Sharing Rule* configuration drawer establishing record sharing from the Service Technician role to the CEO role for the Service_Request_Form Report entity with Read Only access:

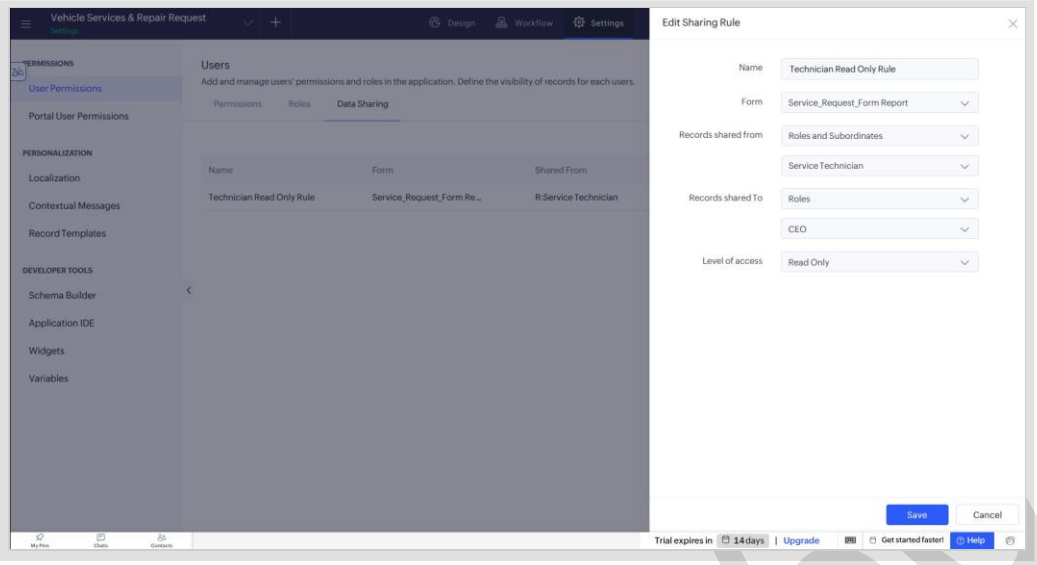


Figure 5: Data Sharing Rule Builder — Configuring "Technician Read Only Rule" for Service_Request_Form Report

3.6 Active Data Sharing Matrix — Enforced Record Access Governance

The Data Sharing rules table view confirming the active deployment of the record-level security rule linking technicians to executive oversight:

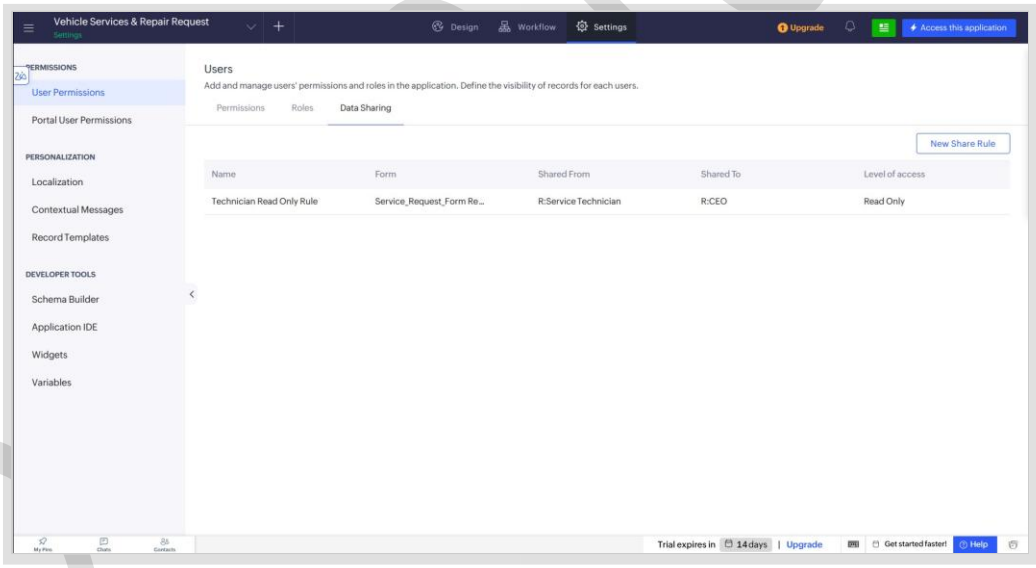


Figure 6: Data Sharing Rules Table — Verified Deployment of "Technician Read Only Rule" (From: R:Service Technician → To: R:CEO)

3.7 Role-Based Access Control (RBAC) Specification Matrix

The table below summarizes the operational responsibilities, data access boundaries, and component-level rights assigned across the application roles:

Role Identifier	Hierarchical Superior	Assigned Permission Profile	Record Scope & Access Boundaries	Operational Responsibility

CEO	*Top-Level Root*	Administrator / Write	Full visibility across all organizational service requests and audit reports.	Executive oversight, resource allocation, and organizational reporting.
Service Manager	CEO	Write	Read and write access to all service intake records, cost estimates, and bay assignments.	Supervision of workshop operations, technician task assignment, and job approvals.
Service Technician	CEO	Restricted Read / Write	Restricted to assigned vehicle service request records; shared read-only to CEO via Data Sharing Rules.	Execution of mechanical diagnostics, repair updates, and service completion notes.

3.8 Data Sharing Rules Governance Specification

The table below details the record-level sharing criteria configured within Zoho Creator to regulate cross-role record visibility:

Rule Name	Target Application Component	Source Entity (Shared From)	Target Entity (Shared To)	Access Level	Governance Objective
Technician Read Only Rule	Service_Request_Form Report	Roles and Subordinates: Service Technician	Roles: CEO	Read Only	Provides organizational leadership with comprehensive audit visibility into technician work logs while preventing accidental record alteration.

4. Result / Observations

User Management and Role-Based Access Control (RBAC) were successfully configured, implemented, and verified in the Zoho Creator **Vehicle Services & Repair Request** application:

1. Default permission profiles (*Read* and *Write*) were reviewed and analyzed to understand component-level security abstractions across forms and reports.
2. A multi-tier organizational role hierarchy was established under the root CEO role, provisioning the Service Manager supervisory role and the Service Technician operational role.
3. Granular record-level security was enforced through the creation and activation of the Technician Read Only Rule in the Data Sharing module, mapping vehicle service report records from technicians to executive oversight.

4. The deployed RBAC architecture successfully adheres to the principle of least privilege, preventing unauthorized record modification while ensuring seamless operational auditability.

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